

Data Science in Economic History: Spatial Analysis and Mapping

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Lecture 1: Introduction to the Course

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Introduction

- This course provides an overview of spatial analysis and mapping in R with applications in economic history.
- **First part:**
 - Introduction to the basic toolkit in geospatial research (as used in economics).
 - Topics: Handling of vector and raster data, processing of spatial data, map making.
- **Second part:**
 - Applies the tools introduced in the first part.
 - Replicates spatial analyses of papers in economic history and long-run development.

Textbooks

The following two online textbooks cover much of the tools discussed in this course:

1. Taro Mieno (2022). R as GIS for Economists. (short: GISEcon)
<https://tmieno2.github.io/R-as-GIS-for-Economists/>
2. Robin Lovelace, Jakub Nowosad, Jannes Muenchow (2023). Geocomputation with R. 2nd edition. CRC Press. (short: LNM)
<https://geocompr.robinlovelace.net/>

Geographic Information Systems (GIS)

- GIS is a computer-based system designed to capture, store, manipulate, analyze, and present spatial data.
- Spatial data is any data that has a location or geographic component, such as GPS coordinates, addresses, or zip codes.
- GIS allows users to visualize spatial data on maps, and to perform spatial analyses and modeling.
- Many of you might have some first experience with using GIS software emphasizing graphical user interfaces (QGIS, ArcGIS).

R as GIS

- This course focuses on programming GIS tasks using R rather than manually implementing them using graphical interfaces.
- Why? Two key advantages of command-line interfaces:
 1. Reproducibility¹ is essential for scientific research.
 2. Manual operations are not scalable.
- Python is a valuable alternative to R.

Why GIS for economists & economic historians?

There are at least two reasons for the increasing popularity of GIS (e.g., Kudamatsu, 2018).

1. **Data:** GIS allows economists to observe the previously unobserved.
2. **Identification:** Geography as a source of exogenous variation.

GIS as data source: Satellite images²

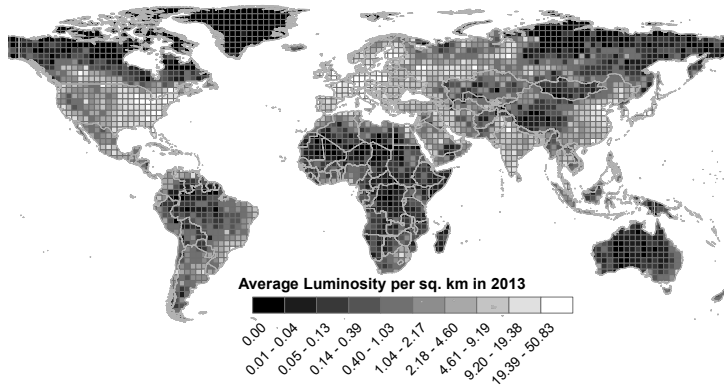
Satellite images can close gaps in the official statistics due to illegality (e.g. deforestation in India) or lack of state capacity (e.g. GDP growth).



Source: Michalopoulos and Papaioannou (2018). *Spatial Patterns of Development*. Fig. 1.

GIS as data source: Satellite images (ctd.)

Satellite images are also not bound to administrative borders and can be aggregated as needed.



Source: Michalopoulos and Papaioannou (2018). *Spatial Patterns of Development*. Fig. 7.

GIS as data source: Old maps

Using GIS, scanned old maps can be used to construct spatial historical data often retained by no other written source.

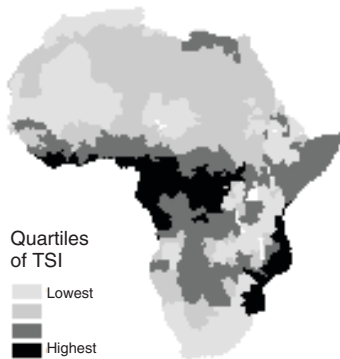


Source: Magnus, O. (1572) *Map of the Sea*. Rome: Antoine Lafréry. <https://www.loc.gov/item/2021668418/>.

GIS for identification: Examples

Climate as a source of (exogeneous?) variation (e.g. TseTse fly suitability index used in Alsan, 2015).

Panel A. TseTse suitability index (1871)



Source: Alsan, M. (2015). *The Effect of the TseTse Fly on African Development*. Fig. 4, Panel A.

GIS for identification: Examples (ctd.)

Geographic features as instrumental variables (e.g. distance to important destinations of slave trades in Nunn, 2008).



FIGURE V
Example Showing the Distance Instruments for Burkina Faso

Source: Nunn, N. (2008). *The Long-Term Effects of Africa's Slave Trades*. Fig. V.

GIS for identification: Examples (ctd.)

Regression discontinuities using (historical) borders (e.g. borders of Peru's forced mining labor system in Dell, 2008).

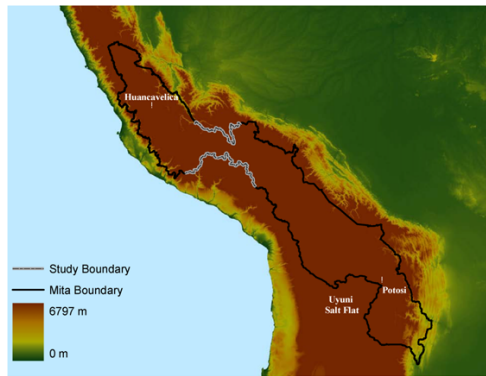


FIGURE 1.—The *mita* boundary is in black and the study boundary in light gray. Districts falling inside the contiguous area formed by the *mita* boundary contributed to the *mita*. Elevation is shown in the background.

Source: Dell, M. (2010). *The Persistent Effects of Peru's Mining Mita*. Fig. 1.

Demonstration I: Nunn (2008)

- Studies the effect of Africa's slave trades 1400-1900 on per capita GDP today.
- Uses an IV approach, instrumenting slaves taken from a country with the distance to the nearest slave trade centre in the Americas.
- Let's see how to construct the instrumental variable using R as GIS...

Instrumental variable

Distance from the point on the coast that is closest to the country's centroid to the closest major market of the Atlantic slave trade.

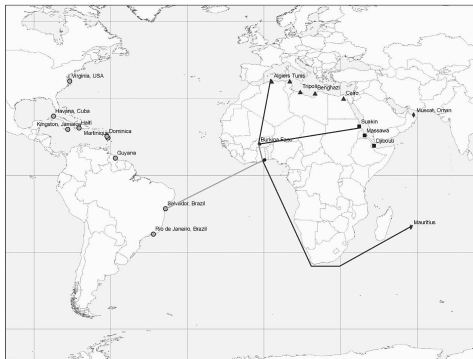


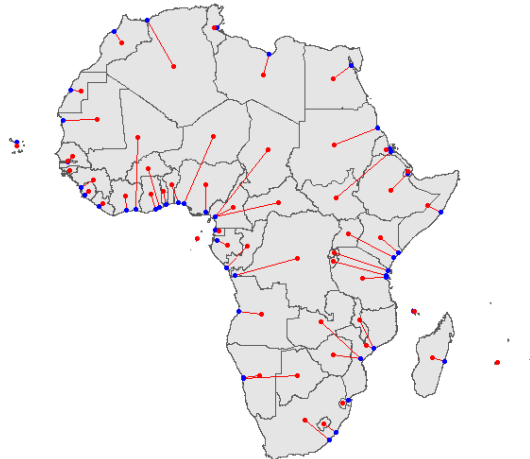
FIGURE V
Example Showing the Distance Instruments for Burkina Faso

Source: Nunn, N. (2008). *The Long-Term Effects of Africa's Slave Trades*. Fig. V.

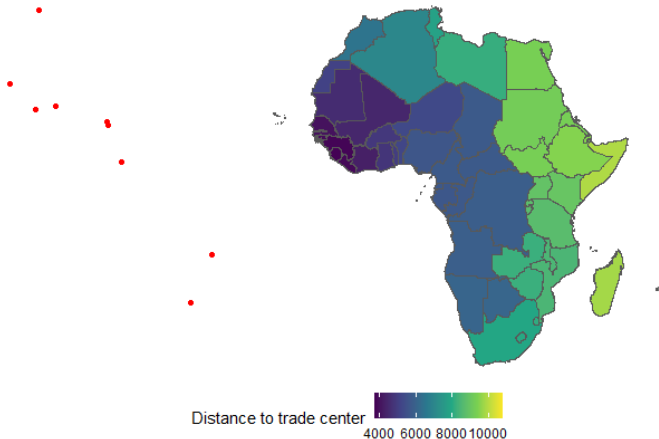
Determine each country's centroid ³



Determine point on the coast closest to centroid ☐



Calculate distance to closest major market

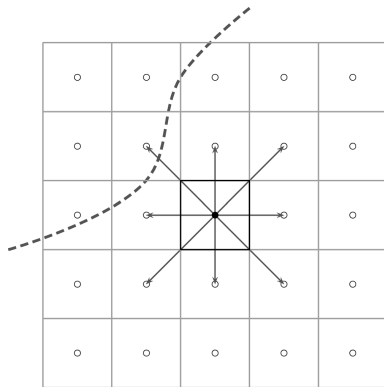


Demonstration II: Nunn and Puga (2012)

- Rugged terrain is tough to farm, costly to traverse, and often inhospitable to live in—and thus impairs economic development.
- Study shows that in Africa, countries with a rugged landscape tend to perform better than flatter ones.
- In Africa, rugged terrain decreased the exposure to the slave trades and thus their negative long-run effects on economic development.
- Let's see how to construct a ruggedness index using R as GIS...

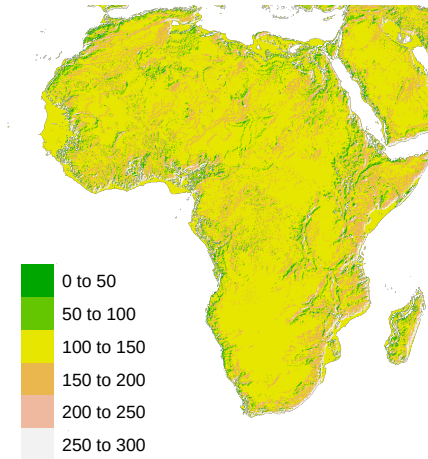
Measuring ruggedness

Square root of the sum of the squared differences in elevation between the central point and the eight adjacent points.⁴



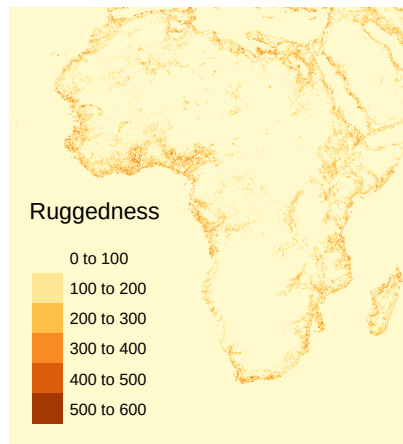
Source: Nunn, N. and D. Puga (2012). *Ruggedness: The Blessing of Bad Geography in Africa*. Fig. 1.

Load and plot raster data on elevation 5



Let R calculate the ruggedness index 🖥️

- Define function for terrain ruggedness index (TRI).
- Loop through raster cells and calculate TRI (see map).
- Averaging across grid cells on land gives a country's index.



Seminar Structure

Date	Topic	Reading
29.04.	Introduction to the Course	GIS Ch. 1
06.05.	Introduction to R	https://github.com/timuroeztuerk/r-lectures
13.05.	Vector Data I	LNМ Ch. 2.2; GIS Ch. 2
20.05.	Vector Data II	LNМ Ch. 3.2, 4.2, 5.2; GIS Ch. 3
27.05.	Raster Data I	LNМ Ch. 2.3; GIS Ch. 4
03.06.	Raster Data II	LNМ Ch. 3.3, 4.3, 5.3; GIS Ch. 5
17.06.	Raster-Vector Interactions	LNМ Ch. 6
24.06.	Making Maps with R	LNМ Ch. 9
01.07.	Application Session I	
08.07.	Application Session II	
15.07.	Project Presentations I	
22.07.	Project Presentations II	

General information

- Slides and R-scripts will be made available on **e-learning** and **GitHub**.
- **Target groups:**
 - Master's students in History and Economics, Economics and related degree programmes
 - Advanced undergraduate students with a profound knowledge of econometrics and R
- **Prerequisites:**
 - Students should be familiar with basic econometric methods.
 - Prior exposure to R is recommended (although not strictly necessary).

Accessing R from the University Computers

R and RStudio is available, you need to request it from the IT department.

https://www.intranet.uni-bayreuth.de/de/intranet_its/raeume-und-ausstattung/PC-Pools/porta/cipinfo.php?ACTION=SOFTWARE

PC-Pools

Softwarefreischaltung

Die verfügbare Software können Sie der [Softwareliste](#) entnehmen.

Um Zugriff auf Kurs-Software zu erhalten, füllen Sie das Formular aus:

Kurs-Software auswählen	▼
Ihr Name	
Ihre Benutzerkennung	
Titel der Lehrveranstaltung	
Dozierende	

Anfrage absenden

Grading

1. Two take-home **problem sets**: 25%

- Problem sets will apply methods taught in the course to real-world data using R.
- Due on 27.05. and 24.06.2025.

2. **Term Paper** (8-10 Pages): (50 + 25)%

- Term paper will analyse a research question with R using the methods taught in the course.
- Involves both coding and discussion of results.
- Due on 30.9.2025.
- Presentations (15-20 minutes) on 15.07. and 22.07.2025. (25% of the grade)

Readings (* optional; ** recommended)

- * Kudamatsu, M. (2018). GIS for Credible Identification Strategies in Economics Research. *CESifo Economic Studies* 64 (2): 327–338.
- * Michalopoulos, S. and E. Papaioannou (2018). Spatial Patterns of Development: A Meso Approach. *Annual Review of Economics* 10 (1): 383–410.
- ** Mieno, T. (2022). *R as GIS for Economists*. Chapter 1.6 and 1.7.

References

Alsan, M. (2015). The Effect of the TseTse Fly on African Development. *American Economic Review* 105 (1): 382–410.

Nunn N. (2008). The Long Term Effects of Africa's Slave Trades. *Quarterly Journal of Economics* 123(1): 139–176.

Nunn, N. and D. Puga (2012). Ruggedness: The Blessing of Bad Geography in Africa. *The Review of Economics and Statistics* 94 (1): 20–36.